Batterjee Medical College
For Science and Technology

Hematological Disorders

Internal Medicine Department

Learning Objectives

Knowledge

- Relate pathophysiological basics and etiology of anaemia
- Describe epidemiology, manifestations, complications and management of anaemia

Skills

- Interpret clinical manifestations and investigations of anaemia to formulate reasonable diagnosis.
- Apply evidence-based management plan for anaemia
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

- Megaloblastic Anemia (B12 & Folate deficiency: Pathogenesis, Neuro manifestations)
- Aplastic Anemia Etiology & Pancytopenia Evaluation
- Hemolytic Anemia Basics: Intravascular vs Extravascular mechanisms
- G6PD Deficiency (Enzymopathy, Oxidative triggers, Heinz bodies, Bite cells)

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Megaloblastic anemia

Vitamin B12 Deficiency Anemia

Physiology of Vit B12

- **Source:** in animal sources only, not in plants
- **Requirements:** 1 ug/day
- **Absorption:** Vit B12 binds to intrinsic factor produced by the parietal cells in the stomach to be absorbed in the terminal part of ileum where it binds to a specific receptor
- **Transport:** in the plasma bound to transcobalamine-II
- **Stores:** in the liver sufficient for 2-3 years of normal consumption
- **Function:**
  - Synthesis of DNA in dividing cells
  - Myelination of nerve fibers

Etiology

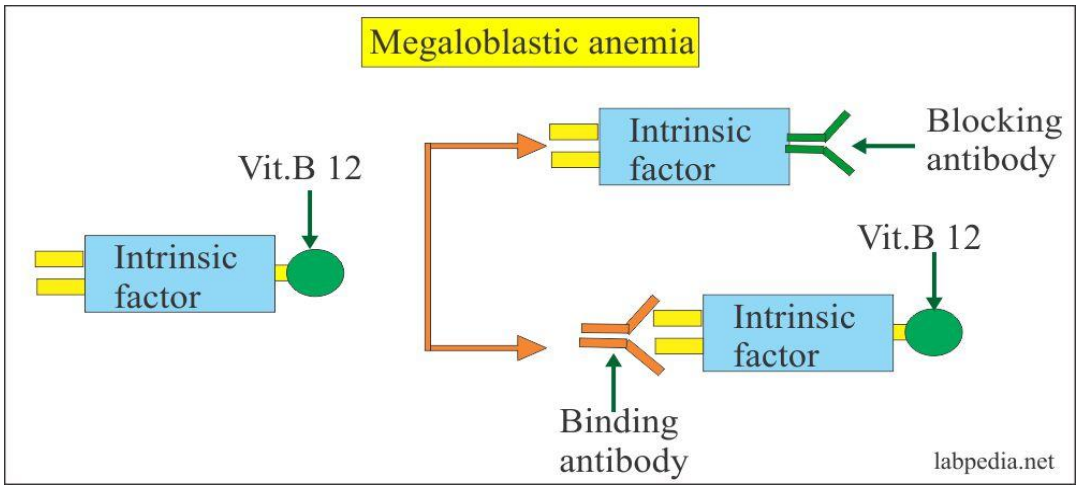
1. **Diminished intake:**
    - Dysphagia, starvation
    - Vegetarians
2. **Diminished absorption:**
    - Gastrectomy, gastritis, gastric carcinoma
    - Addissonian pernicious anemia:

autoimmune disease, which affects north Europe, females> males, rare below age 30 years, The disease is genetic (associated with HLA – A3 & B7). with 3 types of antibodies:

- Anti-parietal cells antibodies
- Blocking antibodies: that prevents binding of Vit B 12 to intrinsic factor

- Binding antibodies: that prevents binding of intrinsic factor-Vit B12 complex to terminal ileum receptors

  It may be associated with other autoimmune disorders e.g. thyroiditis, vitiligo.



- Ileal disease:
  - Regional ileitis, T.B
  - Diphylopotharium latum
  - Malabsorption syndrome

3. **Increase requirements:** Pregnancy, malignancy
4. **Reduced storage:** Liver cirrhosis (uncommon)
5. **Others:** Lack of transcobalamin II (congenital).

Pathogenesis

- In dividing cells there's duplication of DNA & cytoplasm expansion (RNA).
- In case of Vit B12 deficiency, imbalance exists between DNA synthesis (depends on B12) and cytoplasm (RNA is not affected)
  →delay in cell division → formation of megaloblasts.

1. B.M.

Some megalobasts are destroyed in bone marrow (ineffective erythropoiesis):

1. Destruction of RBCs: ↑LDH, ↑serum iron & ↑ hemosiderin in BM.
2. Destruction of platelets → thrombocytopenia.
3. Destruction of WBCs: • serum muramidase.

2.Peripheral blood:

Released cells are at variable stage of development & undergo further destruction in RES → pancytopenia:

a) RBCs: anisocytosis, poikilocytosis, cabot ring · & Howel-Jolly bodies (nuclear remmants).
b) Platelets: More thrombocytopenia.
c) WBCs: Leucopenia with multi segmented PMNLs.

3.GIT: Atrophy of epithelial cells occurs due to rapid turn over of these cells.

4. CNS: Defective myelination of nerve fibers→impaired nerve conduction.

Clinical Picture

1. **C/P of anemia in general**
2. **C/P specific for Vit B12 deficiency:**

➢**Blood**:
- Bleeding tendency (↓ platelets).
- Liability for infection (↓ WBCs).

➢**GIT**:
- Tongue: Red glazed.
- Stomach: Dyspepsia, nusea, vomiting.
- Intestine: Diarrhea.
- Hepatosplenomegaly.

➢**CNS**: Subacute combined degeneration (SCD) → (neurology).

Student pearl

The Spinocerebellar tracts, lateral Corticospinal tracts, and Dorsal columns are affected in Subacute Combined Degeneration.

Clinical pearl

Always consider vitamin B_{12} deficiency when evaluating patients with dementia.

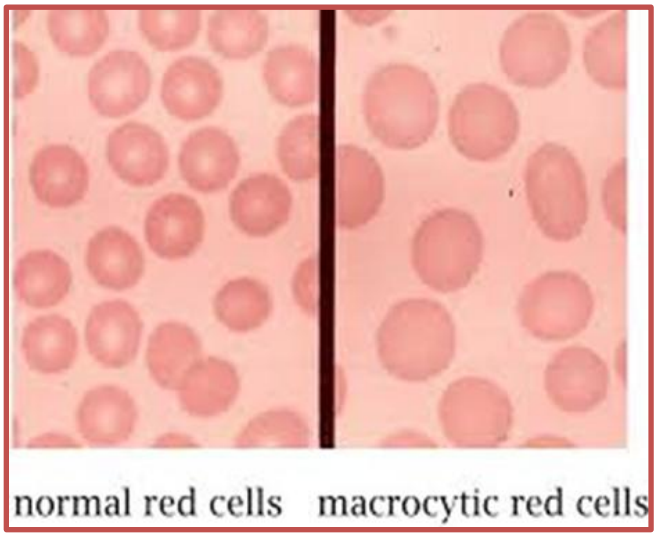
3. **C/P of cause:** e.g. In addission's aneamia (preniciouse anemia)

- Other autoimmune disease e.g. addision, vitiligo, Grave's disease.
- There's increased incidence of gastric carcinoma.

Investigations

I. ↓ Hb, ↓ PCV, ↓ RBCs, ↑ESR.

II. Type of anemia: Megaloblastic normochromic:
- MCV(> 115 FL). - Normal MCHC.



III. The cause:

➢ **Pancytopenia**

1. **RBCs:** Howel-Jolly bodies, cabbot rings, anisocytosis & poikilocytosis +
   Platelets: Thrombocytopenia + • bleeding time.
2. **WBCs:** Leucopenia + • segmented cells + • muramidase.
3. **BM:** -Megaloblast precursors. - Reticulocytes (characterstic).

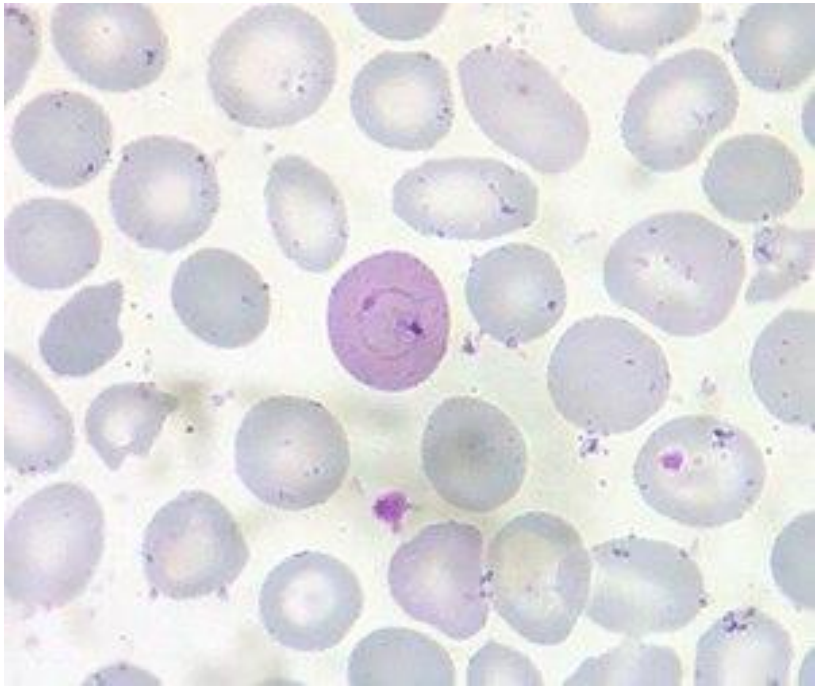
➢ **↓ Vit B12**

1. Low serum Vit B12 (200 – 800 Pg/ml).
2. IM Vit B12 → Reticulocytosis (therapeutic test).

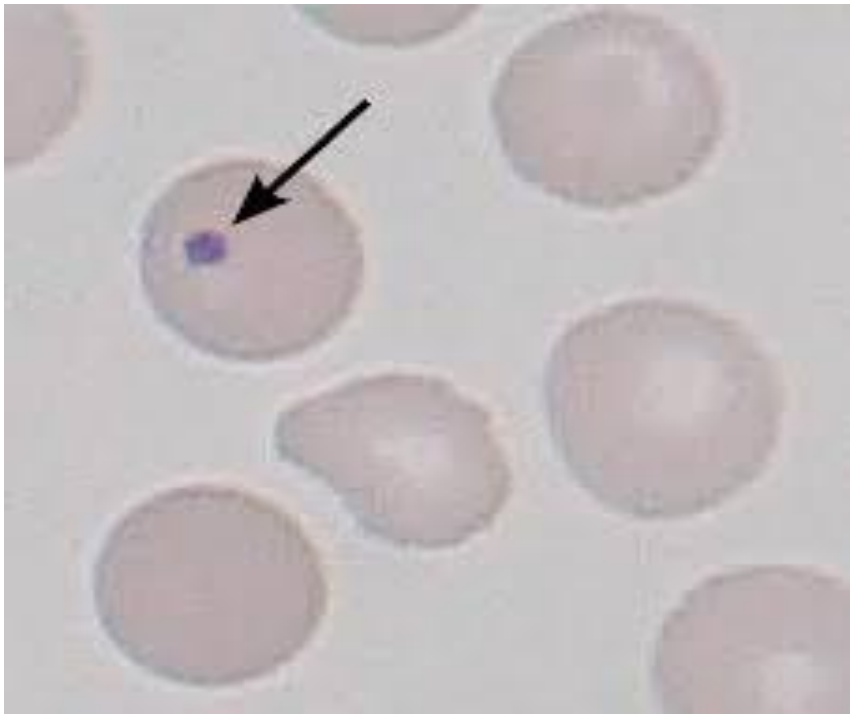
➢ **Methylmalonic acid (MMA):**

↑ Homocysteine + ↑ MMA → B12 deficiency

↑ Homocysteine + normal MMA → Folate deficiency



Cabot rings are thin, red-violet staining, threadlike strands in the shape of a loop or figure-8 that are found on rare occasions in red blood cells (erythrocytes)



- Howell–Jolly bodies are histopathological findings of basophilic nuclear remnants (clusters of DNA) in circulating erythrocytes. During maturation in the bone marrow late erythroblasts normally expel their nuclei, but in some cases a small portion of DNA remains

VI. Investigation of Etiology: e.g.

➢**Preniciouse aneamia:**

- Detection of antibodies.
- Gastric function test for achlorhydria.
- Endoscopy to exclude gastric carcinoma.

➢**Investigations for malabsorption syndrome.**

- **Shilling test:**
  - **Procedure**: 1000 ug B12 IM → Then 1 Ug oral radioactive B12.
  - **Normally**: 25% of radioactive Vit B12 appears in urine / 24 hrs
  - **Abnormal response**: < 5% of radioactive B12 / 24 hr urine → Malabsorption.

	Description	Radiolabeled vitamin B_{12} levels in urine
Stage 1	• Oral radiolabeled vitamin B_{12} PLUS IM dose of unlabeled vitamin B_{12}	• Normal excretion: Insufficient dietary intake or an increased demand is the most likely cause. • Excretion below what is expected: Perform stage 2.
Stage 2	• Oral radiolabeled vitamin B_{12} PLUS intrinsic factor	• Normal excretion: Malabsorption due to lack of intrinsic factor is the most likely cause. • Excretion remains below expected: Perform stage 3.
Stage 3	• Oral radiolabeled vitamin B_{12} PLUS antibiotics	• Normal excretion indicates bacterial overgrowth. • Excretion remains below expected: Perform stage 4.
Stage 4	• Oral radiolabeled vitamin B_{12} PLUS pancreatic enzymes	• Normal excretion indicates pancreatic enzyme deficiency (e.g., chronic pancreatitis).

Treatment

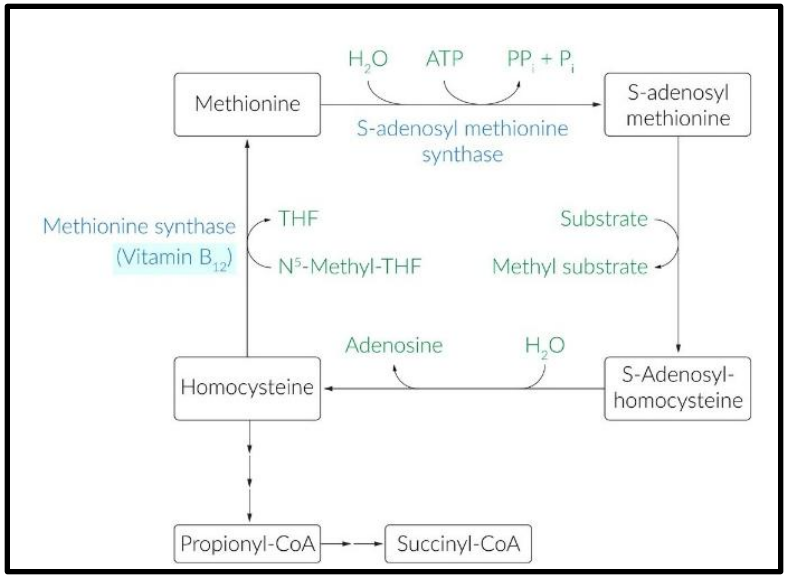
I. Replacement therapy: B12 (Hydroxy cobalamine) 100 Ug IM daily till RBCs count to 5 million /mm3 one ampoule IM/ month for life in cases of malabsorption.

II. Treatment of cause

III. Blood transfusion if:

- HB < 7 gm%.
- HF or ischemic heart disease.

Folic Acid Deficiency



Physiology

- **Source:** In vegetables.
- **Requirements:** 50 ug / day.
- **Absorption:** Intestine.
- **Stores**: In liver 5-15 mg (sufficient for only 2-3 months).
- **Function:** For DNA synthesis.

➢**Vit B12 and folic acid are needed for methylation of homocystiene to methionine which is needed for conversion of uracil to thymidine to be incorporated in DNA**

Etiology

1) ↓ **Intake:** anorexia, alcoholism, lack of vegetable intake.

2) ↓**Absorption:**

   - Malabsorption syndrome.
   - Drugs: Anti convulsants, methotrexate.

3) ↓**Storage:** Liver cirrhosis.

4) ↑ **Requirements:**

   - Pregnancy.
   - Malignancy.
   - Hemodialysis.
   - Hemolytic aneamia.

5) ↑ **Utilization:** e.g. methotrexate & trimiethoprim.

Clinical Picture

As Vit B12 except CNS manifestations which are not present in folic acid deficiency.

Investigations

1) ↓Hb, ↓ PCV, ↓RBCs, ↑ESR.

2) Type: As B12.

3) Of cause: as B12 but:

- Low folic acid level: (N = 6 – 20 ngml/ml).
- Homocysteine level
- FIGLU test (historical, not used nowadays): Normal: Histidine → formiminoglutamic acid (FIGLU) → glutamic acid.
- In folic acid deficiency anemia → ↑FIGLU in urine.

4) Underlying etiology: e.g. investigation for malabsorption syndrome.

Treatment

1.Replacement: Folic acid 5-15 mg orally/day.

2.Of cause: e.g. stop methotrexate.

3.Blood transfusion.

Student pearl

Folic acid supplementation is recommended for all individuals planning or capable of pregnancy because of the risk of neural tube defects due to maternal folate deficiency. [12]

NB: Empirical treatment by folic acid in patients with Vit B12 deficiency will exacerbate CNS manifestations due to consumption of already low Vit B12 in activating folic acid.

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Bone Marrow Failure (Aplastic Anemia)

Types

➢ **Unipotent:** Affects one element of blood

1- RBCs:

- ↓ ERP:
  - Renal failure.
  - Oat cell carcinoma(small cell lung cancer) → Secrete anti-enythropoietein
  - Thymoma

- A plastic crisis of hemolytic anemias.

2. Platelets: - TAR syndrome: thrombocytopenia + absent radii

3. WBCs: - Idiopathic. - Congenital.

➢**Pleuripotent BM failure:** Affects all blood elements Failure of development of BM stem cells → Pancytopenia BM, & empty blood).

Aetiology

1. **Idiopathic:** (50% of cases) due to suppression of BM by cytotoxic T-cell (autoimmune).
2. **Irradiation** e.g. radioactive phosphorus
3. **Drugs:** Anti's:
   - Thyroid: Neomercazole.
   - Emetic: Clorpropamide.
   - Antibiotics: Chloramphenicol.
4. **Chemicals:** benzene, insecticide.
5. **Infections:** Dissiminated TB, EBV, hepatitis.
6. **Immune:** SLE.
7. **B.M. replacement** by leukemia, lymphoma(myetophthitic aneamia).

Clinical Picture:

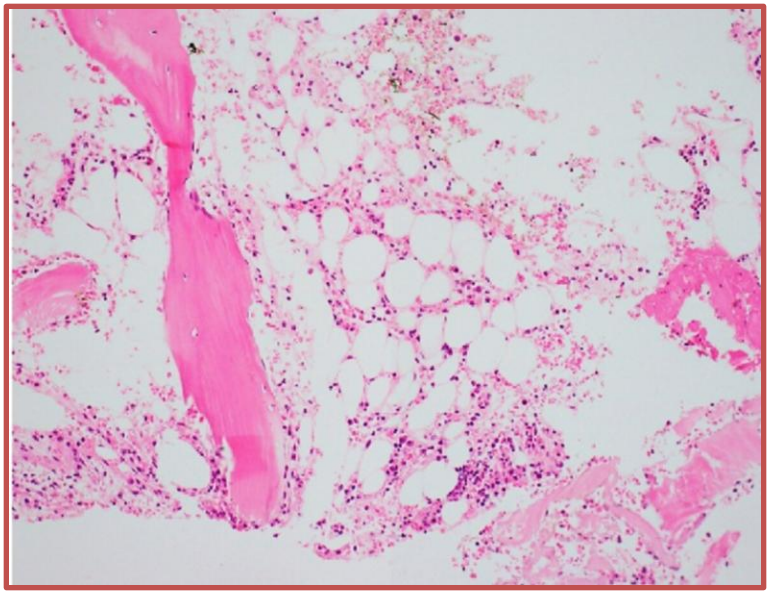
1. **General features of anemia**

2. **Specific features:**

   - ↓ Platelets: purpura, bleeding tendency.
   - ↓ WBCs: ↑ Liability for infection especially with fungi or opportunistic organisms. It affect areas already inhabitant of bacteria (e.g. mouth) & there's no pus cells.

3. **Of cause** e.g. skeletal deformities.

Investigation



1. **For all anemia:** ¯ Hb, HCT, RBCs & • ESR
2. **Type of anemia:** Normocytic normochromic
   - MCV, MCH normal.
3. **Causes**:
   - ↓platelets & WBCs.
   - Absent reticulocytes (most specific).
4. **Aetiology:** Bone narrow biopsy shows hypoplastic narrow

Differential diagnosis: Causes of pancytopenia:

- Hypersplenism
- Megaloplastic anemia
- Aplastic anemia

Treatment

1. Supportive treatment:

- Anemia → Packed RBCs.
- Thrombocytopenia→ Platelet
- Avoid trauma (Shaving, IM injection).
- Antibiotics→ (imipienam) + antifugal + Antiviral

2. BM stimulation:

- Steroid
- Androgen Stimulate RBCs
- GM-CSF ↑WBCs
- Lithium

3. Treatment of Cause:

- Idiopathic type: Immunosuppressant as: Prednisone.
- Stop causative drug.

3. BM transplantation

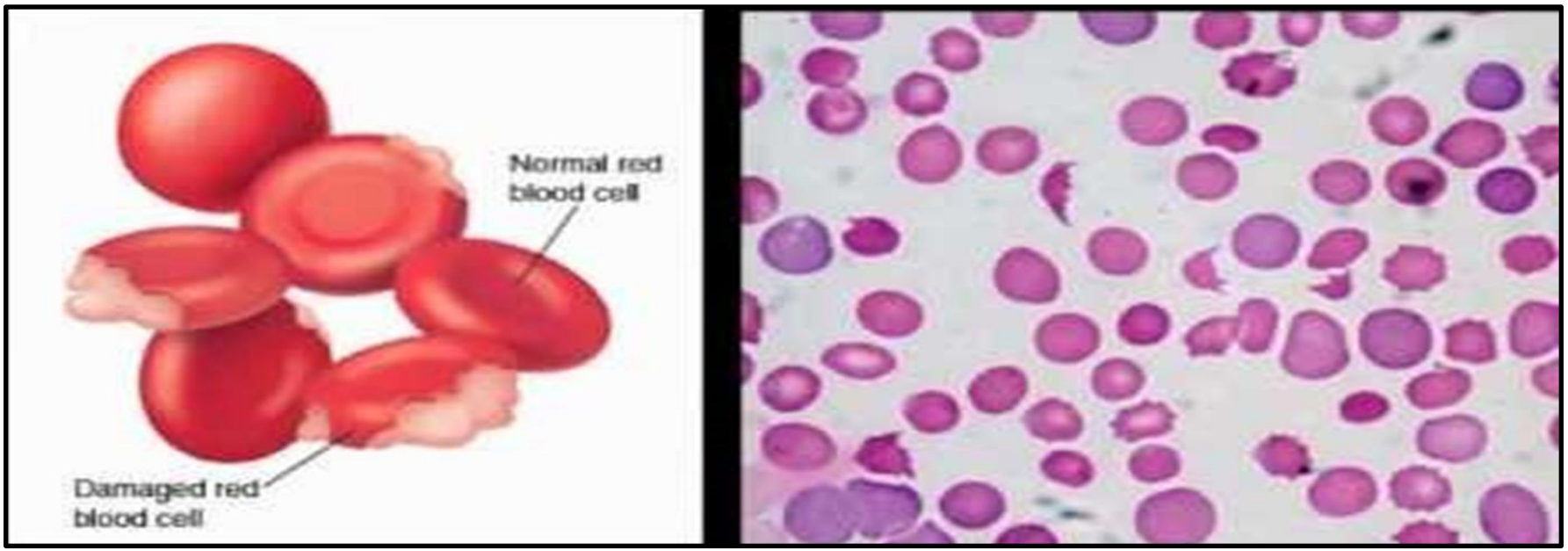
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Hemolytic Anemia

Definition: **Shortening of RBCs Life Span**

Physiology

- The normal life span of RBCs is 120 days.
- The bone marrow can compensate for shortening of this life span of RBCs by increasing their production.
- When life span of RBCs is shortened to 1/6 – 1/8 the normal (15-20 days), anemia starts to manifest (uncompensated hemolytic anemia).
- Hemolysis occurs in the RES in 90% of cases and intra-vascular in 10% of cases.



Etiology Hemolytic anemia occurs either due to:

1. Corpuscular defect:

A. Cell wall defect:

- Hereditary spherocytosis.
- Hereditary eliptocytosis.
- PNH

B. Enzymatic defect:

- Glucose – 6 – phosphate dehydrogenase deficiency (G6PD).
- Pyruvate kinase deficiency.

C. Hemoglobin defect:

- Thallassemia.
- Sickle cell anemia.

2. Extracorpuscular causes:

A. Traumatic(mechanical):

- Prosthetic valves or sclero-calcific aortic valve.
- microangiopathic hemolytic anemia:

1. DIC 2. Hemolytic uremic syndrome 3. MalignantHTN
4. Scleroderma 5. Thrombotic thrombocytopenic purpura

B. Hypersplenism.

C. Immunological:

- Autoimmune: Cold or warm hemolytic anemia.
- Alloimmune: Incompatible blood transfusion, RH incompatibility.

D. Infective: Sepsis, malaria, and hemolytic streptococci.

E. Toxic causes: Snake or spider venom, lead poisoning, staphylococcus and copper.

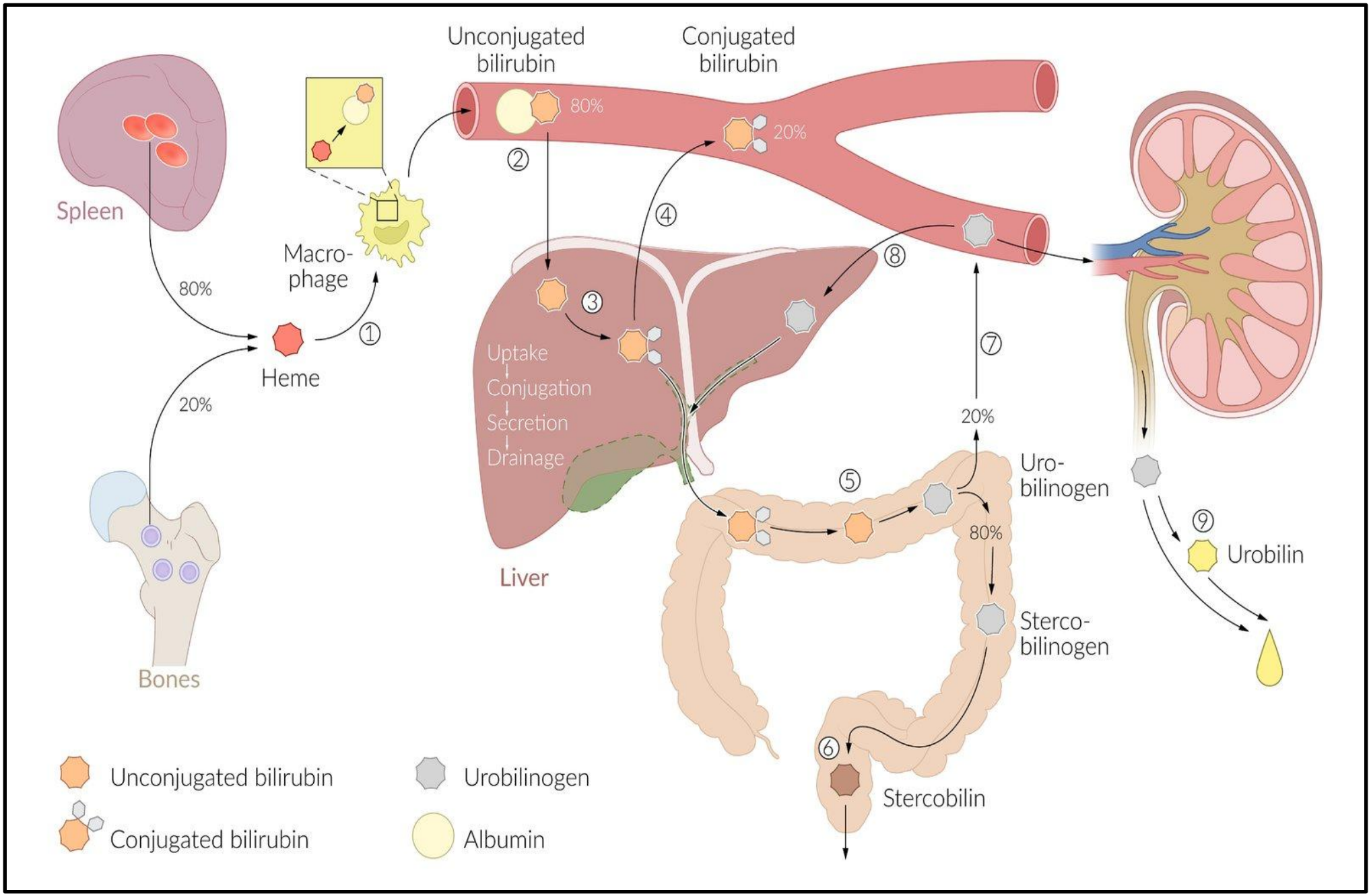
Clinical Picture

I. Of anemia : when hemolysis is uncompensated**.**

II. Of hemolytic anemia.

- **Extravascular hemolysis:**
  - Bone: Expansion of bone ends leading to Mongoloid features.
  - Hepatosplenomegaly: Due to extramedullary hematopoiesis, & hemolysis, hemosidrosis.
  - Jaundice (liver can increase conjugation 8 times and jaundice will occur only in severe hemolysis) & gall bladder stones.
  - Hemosedrosis:
    - Cardiomyopathy.
    - Bronze diabetes.
    - Skin ulcers.
    - Interstitial lung fibrosis.
    - Testicular atrophy.
    - Renal tubular defects.
  - Dark stool and urine darkens on standing.




Unconjugated bilirubin
Conjugated bilirubin
80%
20%
Spleen
Macro-phage
80%
Heme
20%
Bones
Uptake
Conjugation
Secretion
Drainage
Liver
20%
Uro-bilinogen
80%
Sterco-bilinogen
Stercobilin
Urobilin
Unconjugated bilirubin
Conjugated bilirubin
Urobilinogen
Albumin

- Crises:

1. Hemolytic crisis: Associated by increase pallor and jaundice, along with reduction in RBCs and increase in serum billirubin.
2. Aplastic crisis:
    - Caused by parvovirus infection.
    - Causes increase in pallor but without deepening in jaundice or increase in billirubin serum level.
3. Megaloblastic crisis:
    - Due to depletion of Vit B12 and folic acid.
    - Associated with pancytopenia, neurological and GIT manifestations.

- **Intra-vascular hemolysis:**
  - Hemolysis in the vascular stream leads to release of free HB, which is filtered by the kidney leading to hemoglobinemia and hemoglobinuria.
  - This occurs after the binding protein system has been exhausted. This system involves:
    - Haptoglobin.
    - Hempexin. (for research purposes)
    - Albumin.

III. C/P of the cause.

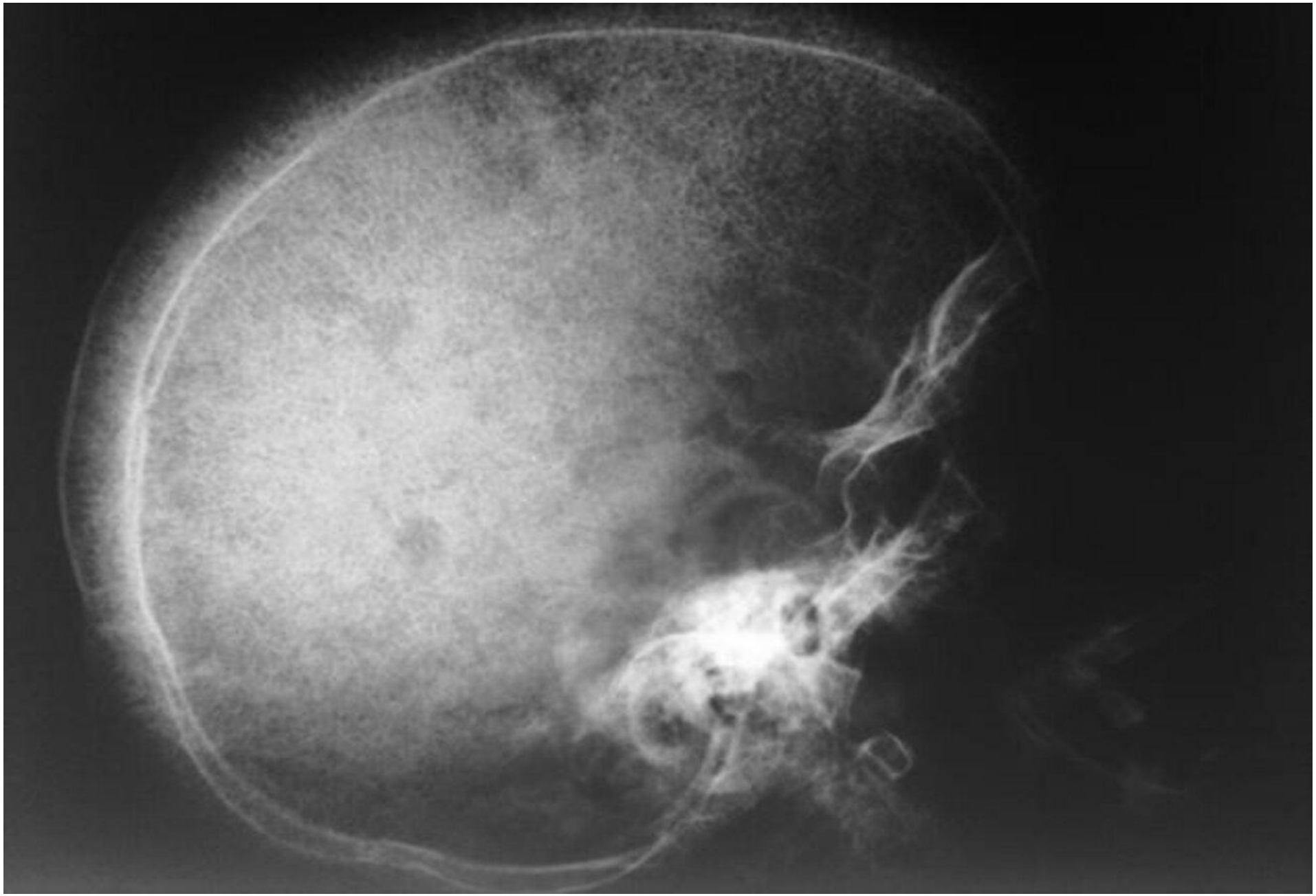
Investigations

1- As all types of anemia: ↓ HB, RBCs, HCT, ↑ ESR.

2- **Type of anemia**: mostly Normocytic normochromic anemia (normal HCV, MCH).

3 For hemolytic anemia:

- Shortening of life span of RBCs as detected by Tc-99 labeled RBCs.
- ↑ LDH serum level.
- BM expansion by x-ray (widening of medulla and thinning of cortex & hair on end appearance).
- ↑ Total billirubin (mainly indirect), ↑ urobilinogen in urine and strocobilinogen in stool.
- ↑ Serum iron and ferritin + ↓ total iron – binding capacity.
- Hepatosplenomegaly and GB stones by sonography.
- WBCs & platelets are increased(hyperactive BM)
- Reticulocytosis



- In intravascular hemolysis there is:
  - Decrease haptoglobin.
  - Decreased hempexin.

> **Clinical pearl**
>
> Haptoglobin levels can be low in both intravascular hemolysis and extravascular hemolysis and, therefore, should not be used to differentiate between the two. [11]

4. Investigation of the cause.

❖ Treatment

1. Blood tansufusion.
2. Folic acid and Vit B12 replacement.
3. Iron chelating agent: Desferroxamine SC.
4. Splenectomy.
5. Treatment of the cause.

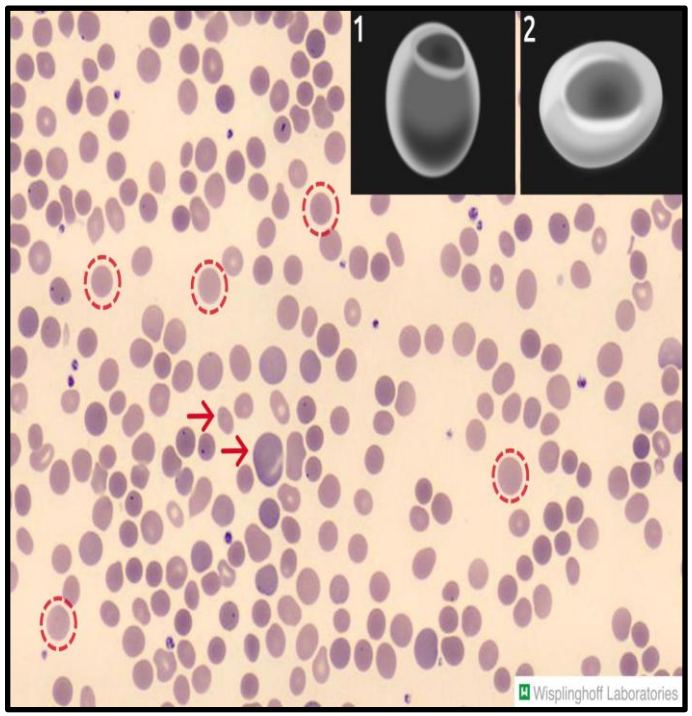
Hereditary Spherocytosis

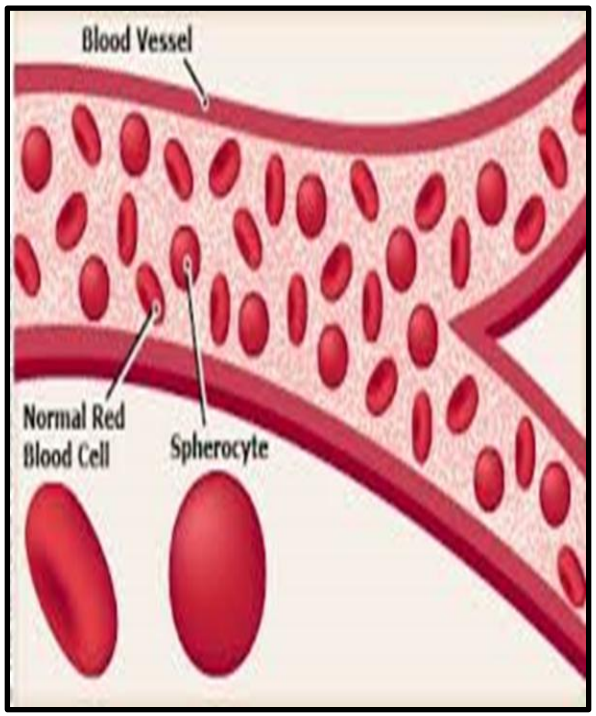
Etiology:

- Hereditary (autosomal dominant)
- There is a defect in the protein composition in RBCs cell wall (ankyrin & spectrin) that makes them more liable to hemolysis in the RES& leads to reduction of RBCs size which become small and spherical in shape(microspherocytes) making them more liable to osmotic rupture.

Clinical Picture:

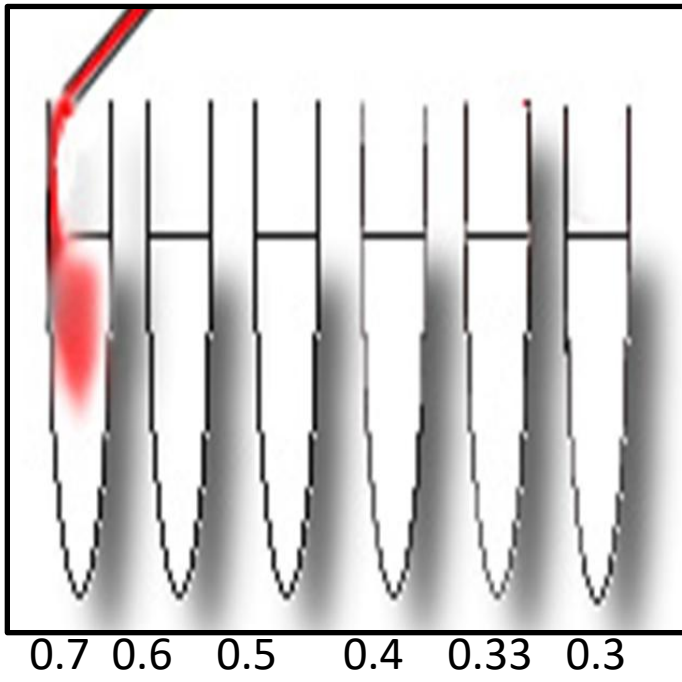
1. As anemia in general
2. As hemolytic anemia
3. Positive family history

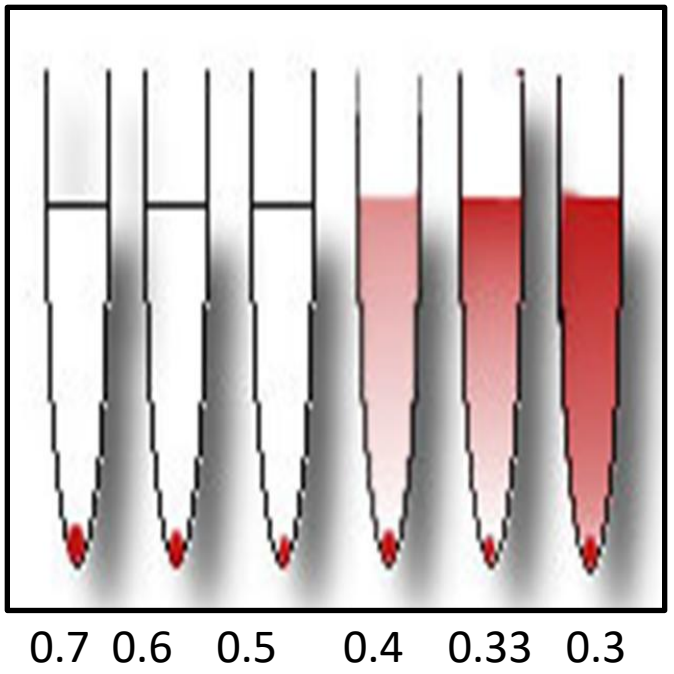




Investigations:

1. For anemia & hemolytic anemia.
2. Type: micorcytic hyperchromic
3. Specific: spherocytes
4. Of cause: Osmotic fragility test: adding RBCs to different dilutions of NaCl (normally hemolysis starts at 0.4 and completed at 0.33, but in spherocytosis it starts at 0.6).
5. EMA binding test (Eosin-5'-maleimide flow cytometry) It is the most sensitive and specific





Treatment

1. As general
2. Splenectomy(better after age of 4) in patients with:

- High transfusion requirements
- Family history of death due to sherocytosis
- Hypersplenism

(G6PD deficiency)

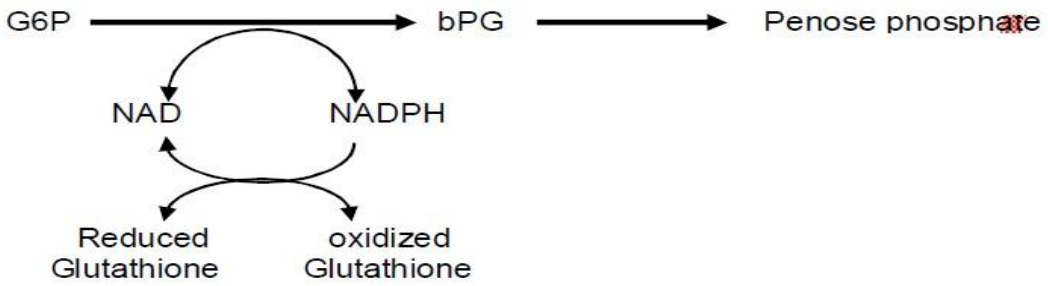
Etiology:

X-linked disease:

- Affect males.
- Females are carriers.

Pathogenesis:

- G6PD enzyme is the 1st step in hexose monophosphate shunt (HMP).
- The shunt is responsible for regenerating reduced Glutathione from oxidized Glutathione by NADPH (H+ donor).
- Reduced glutathione is the 1st protective line against oxidation of Hb to met HB.
- ↓G6PD → oxidation of SH groups in RBCs membrane
  → denaturation of Hb which attaches to RBCs cell membrane (Heinz bodies)
  → destruction of these RBCs in RES



Clinical Picture:

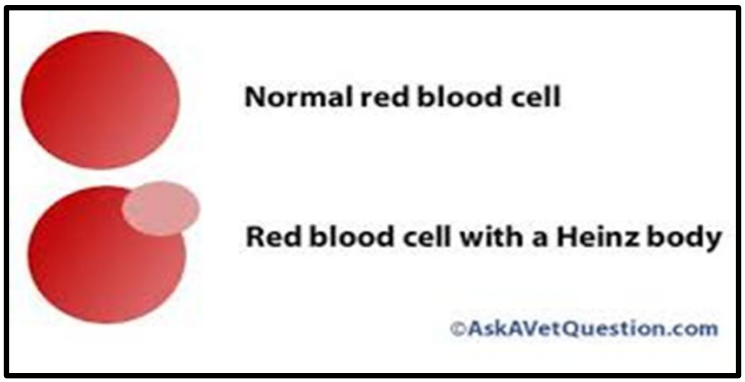
- Hemolysis is precipitated by oxidizing agents:
  - 1. Infection.
  - 2. Drugs:

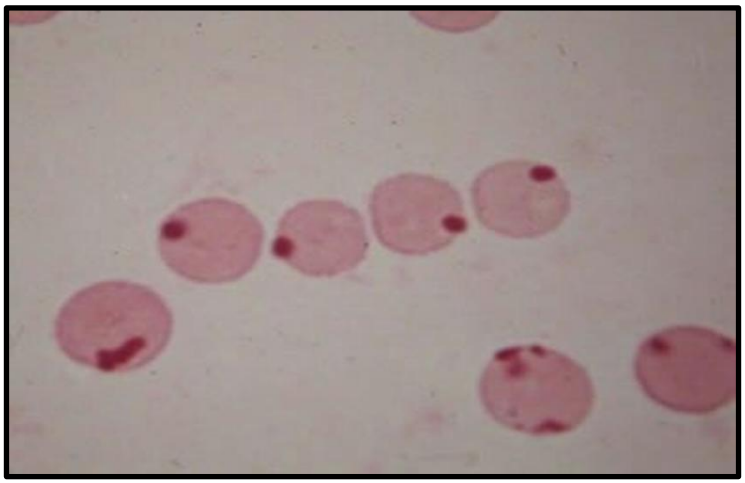
    - Antimalarial. - Antituburculous. - Sulpha. - Antipyreties: Aspirin. - Synthetic Vit K.
  - Metabolic acidosis e.g. DKA
  - Uncooked fava beans (favism).

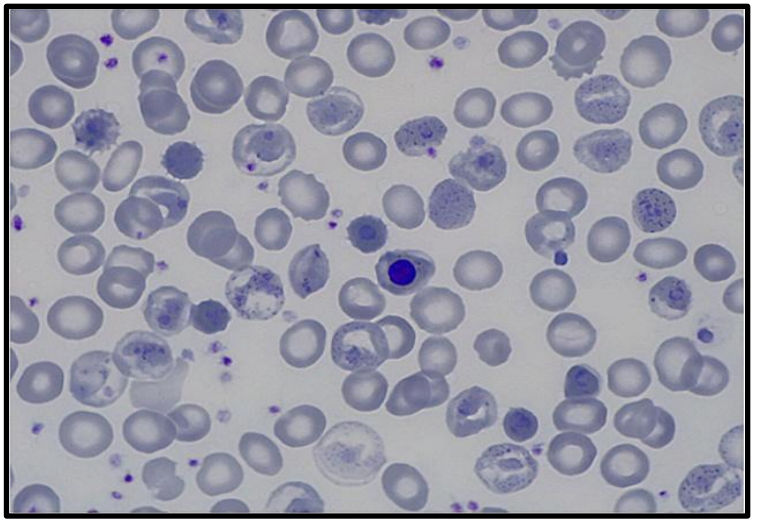
1. As anemia
2. As hemolytic anemia.
3. Specific for cause:
   - **G6PD in negros:** enzyme deficiency in the aging RBCs with mild anemia
   - **G6PD Mediterranean:** enzyme defect in young RBCs with severe hemolysis.
   - **Favism:** there should two genes defective:
     - G6PD gene
     - Gene defect that leads to release of O2 free radicals with metabolism of Fava beans leading to RBCs hemolysis

Investigations:

1. As anemia: ↓ Hb, PCV & RBCs, ↑ ESR.
2. As hemolytic anemia
3. Specific
   - Heinz bodies seen during the attack (stained with methyl violet).
   - ↓ G6PD activity in RBCs: done only 2 months after the last haemolytic attack as the newly generated RBCs during crises have normal level of G6PD.







Treatment:

- In crisis → blood transfusion.
- A void cause → treatment of infection & stop causative drug

Case scenario :

A 44-year-old woman is brought to the emergency department because of confusion and agitation. She was brought by police after she was found walking along a highway. The patient's brother comes to join her soon after her arrival. He says she has peptic ulcer disease and hypertension. He thinks she drinks around half a bottle of vodka daily. Her current medications include omeprazole and hydrochlorothiazide, although the brother is unsure if she takes them regularly. Her temperature is 37.1°C (98.7°F), pulse is 90/min, respirations are 16/min, and blood pressure is 135/90 mm Hg. On mental status examination, she is confused and not oriented to person, place, or time. Neurologic examination shows horizontal nystagmus. Her gait is wide-based with small steps. Her hemoglobin concentration is 9 g/dL. Her serum homocysteine concentration is elevated and her methylmalonic acid concentration is within the reference range. A peripheral blood smear shows hypersegmented neutrophils.

Which of the following is the most likely cause of this patient's anemia?

A Folate deficiency

B Vitamin E deficiency

C Iron deficiency

D Vitamin B6 deficiency

E Vitamin B1 deficiency

F Vitamin B12 deficency

Summary



References

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3. Davidson's Principles & Practice of Medicine, 25th edition (ISBN-13: 978-0443287626)
4. Harrison's Principles of Internal Medicine, 22nd edition (ISBN-13: 978-1265979317)
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Thank You